

Abhijeet Y. Vishwasrao

PhD Researcher · Generative AI and Foundation Models for Physics Simulation

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SUMMARY

PhD researcher building generative and foundation models for physics simulation. Works with diffusion models, flow matching and JEPA latent predictors, coupled with LLM-agent systems, to shorten engineering design cycles and speed up scientific discovery. Has generated terabyte-scale datasets with open-source CFD solvers to train 100M+ parameter models on multi-GPU HPC clusters, and has published in **Nature Communications**, **NeurIPS** and the **Journal of Fluid Mechanics**. Leads collaborations across industry (**NVIDIA**, **RTX**, **CoreWeave**, **PhysicsX**) and academia (Stanford, University of Washington, Caltech).

EXPERIENCE

Graduate Researcher, University of Michigan, Ann Arbor (advisor Prof. R. Vinuesa) Aug 2025 – Present

- Building latent foundation models for parametrized PDE simulations and an agentic explorer (preprint 2026, [arXiv:2604.09584](#)), a multi-LLM-agent system that runs them autonomously for scientific exploration
- Conceived and led Policy-DRIFT (**NeurIPS 2026**, [arXiv:2605.14022](#)), first successful attempt to introduce generative AI in flow control, mentoring a master's student through implementation to publication
- Co-developed AeroJEPA (preprint 2026, [arXiv:2605.05586](#)), a joint-embedding model for 3D aerodynamic fields, now in **NVIDIA PhysicsNeMo**
- Partnering with **RTX (Pratt & Whitney)** on film-cooling flow prediction, and with **CoreWeave** and **PhysicsX** on a JEPA latent foundation model for automotive aerodynamics (DrivAerML, DrivAerNet++), on a CoreWeave grant of ~41,000 A100 GPU-hours

Research Aide, Argonne National Laboratory, Lemont IL (advisors Dr. M. Min, Prof. P. Fischer) Jun 2026 – Aug 2026

- Ran NekRS conjugate heat transfer simulations over $Re = 2500-15500$ for wire-coil inserts in reactor heat exchangers, building the dataset to accelerate these simulations with latent generative models
- Built the two-stage Reynolds-decomposition decoder that recovers the turbulent fluctuations a plain autoencoder loses, with a latent predictor for unseen parameters

Doctoral Researcher (MSCA Fellow), KTH Royal Institute of Technology (advisors Profs. R. Vinuesa, H. Azizpour) Mar 2023 – Aug 2025

- Developed Diff-SPORT (**Nature Communications** 2026, [arXiv:2506.00214](#)), a diffusion-based framework for sparse reconstruction and optimal sensor placement in urban flows, validated on DNS and on experimental data of a scaled-down Beijing neighborhood
- Integrated 3D EDM-class diffusion models into **NVIDIA PhysicsNeMo** and scaled training across multi-GPU nodes
- Ran Nek5000 urban-flow simulations (5+ TB of data) and built an automated pipeline for parametrized 2D-DNS datasets

Visiting Researcher, Center for Turbulence Research, Stanford University (mentor Prof. B. J. McKeon) Jun 2024 – Jul 2024

- Authored the successful proposal to the biennial CTR Summer Program, worked on diffusion-based methods for urban flows

Master's Thesis, CEMEF, MinesParisTech, Université PSL (advisor Prof. E. Hachem) Apr 2022 – Oct 2022

- Deep RL for uniform cooling in quenching (**Int. J. Numer. Methods Heat & Fluid Flow** 2024, [doi:10.1108/HFF-11-2023-0713](#))

Research Trainee, CSIR-National Aerospace Laboratories, Bengaluru (mentor Dr. P. Deshpande) Jul 2019 – Jul 2020

- PIV and unsteady-force experiments on thrust from flexible pitching panels (**J. Fluid Mech.** 2021, [doi:10.1017/jfm.2021.494](#))

SKILLS

ML & generative models: diffusion (EDM), flow matching, VAEs, JEPA, deep RL, surrogate modeling, LLM agents, SHAP

Programming & frameworks: Python, C++, Fortran, CUDA, PyTorch, NVIDIA PhysicsNeMo, Hydra, Docker, vLLM, AutoGen, LangChain

Solvers & HPC: Nek5000, NekRS, OpenFOAM, Cimlib, Basilisk, ANSYS Fluent; SLURM, multi-GPU training on EU and US clusters

EDUCATION

University of Michigan, PhD in Aerospace Engineering, GPA 4.0 (transferred with advisor from KTH) 2025 – May 2027 (exp.)

KTH Royal Institute of Technology, PhD studies in Engineering Mechanics, GPA 4.0 2023 – 2025

École Polytechnique, MSc in Fluid Mechanics 2021 – 2022

Defence Institute of Advanced Technology, Pune, M.Tech in Aerospace Engineering, First Class with Distinction 2018 – 2020

Savitribai Phule Pune University, BE in Mechanical Engineering, First Class with Distinction 2014 – 2018

SELECTED PUBLICATIONS

- Vishwasrao, A.** et al. Diff-SPORT: diffusion-based sensor placement and reconstruction. *Nat. Commun.* 2026. [arXiv:2506.00214](#)
- Mahajan, A.*; **Vishwasrao, A.*** et al. Policy-DRIFT: generative flow control. *NeurIPS* 2026 (*equal contribution). [arXiv:2605.14022](#)
- Giral, F.; **Vishwasrao, A.** et al. AeroJEPA: latent representations for 3D aerodynamic fields. [arXiv:2605.05586](#), 2026
- Deshpande, P.; **Vishwasrao, A.** Kinematics and dynamics of pitching flexible panels. *J. Fluid Mech.* 2021. [doi:10.1017/jfm.2021.494](#)